EXTENDED MONTGOMERY REDUCTION IN n^2+1 DIGIT MULTIPLICATIONS AND A BARRETT–MONTGOMERY DUALITY

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ABSTRACT. Let $p > 1$ be odd and let $R = r^n$ with $r = 2^w$ coprime to p. We exhibit a structural duality between Barrett and Montgomery modular reductions: writing a $2n$-digit integer $c = c_1 R + c_0$, Barrett reduction assembles $c \bmod p$ from a reduction of $c_1 R$ and the limb c_0, whereas Montgomery reduction assembles $cR^{-1} \bmod p$ from a reduction of $c_0 R^{-1}$ and the limb c_1. Standard operand-scanning implementations of either method require $n^2 + n$ digit multiplications. We introduce the *extended Montgomery reduction*, which uses only $n^2 + 1$ digit multiplications by replacing the first $n - 1$ reduction steps with multiplications by the precomputed constant $\rho_1 = r^{-1} \bmod p$. A dual $n^2 + 1$ extended Barrett reduction is obtained by the same principle. Two further variants, a fully parallel form and a logarithmic form, reduce limbs in fewer rounds at the same digit cost.

1. Introduction

Modular multiplication in $\mathbb{Z}/p\mathbb{Z}$ for large unstructured primes p underlies RSA, elliptic-curve cryptography, and polynomial protocols over prime fields. Given a full product $c = ab$ with $0 \le c < R^2$, where $R > p$ is a power of two, one must reduce c modulo p. Division by p is avoided by precomputation: Barrett [1] estimates the quotient from a scaled reciprocal of the modulus, whereas Montgomery [4] represents residues in the coset $\mathbb{F}_p R$ and clears each low digit using a precomputed inverse of the modulus with respect to the digit base. For multi-precision operands with $R = r^n$, the best operand-scanning Montgomery reduction currently known costs $n^2 + n$ digit multiplications and at most as many additions as Montgomery's original loop.

In this note we expand and formalize material first presented in [2]. Section 2 presents the Barrett–Montgomery duality from the product representation $c = c_1 R + c_0$. Section 3 recalls Montgomery's operand-scanning reduction [4] and the dual Barrett scan of [2], each costing $n^2 + n$ digit multiplications. Section 4 introduces *extended Montgomery reduction* (EMR), which needs only $n^2 + 1$ digit multiplications, a saving of $n - 1$ multiplications per reduction over the standard method. Section 5 gives parallel and logarithmic EMR variants at the same digit cost, and notes the dual extended Barrett reduction (EBR). Section 6 bounds the normalization cost of each variant. Section 7 discusses implementation. Section 8 outlines applications to NTT, constant-matrix hashes, and projective elliptic-curve addition.

Throughout, p is an odd positive integer, $r = 2^w$ is the digit base, $R = r^n$, and $n \ge 2$. Digits are written in base r; for $x = \sum_{i=0}^{k-1} x_i r^i$ we write $x = (x_{k-1}, \dots, x_0)_r$. We assume $0 < p < R$ and $\gcd(r, p) = 1$. The duality and digit-count results hold for any input $0 \le c < R^2$; the overflow and normalization bounds of Section 6 assume $0 \le c < pR$, as holds for a product of residues.

2. The Barrett–Montgomery duality

Let c satisfy $0 \le c < R^2$. Define limbs $c_0 = c \bmod R$ and $c_1 = \lfloor c/R \rfloor$, so that

$$c = c_1 R + c_0, \qquad 0 \le c_0 < R, \quad 0 \le c_1 < R. \tag{1}$$

Theorem 2.1 (Barrett–Montgomery duality)**.** *For every integer c with $0 \le c < R^2$,*

$$c \bmod p \equiv \left(c_1 R \bmod p\right) + c_0 \pmod{p}, \tag{2}$$

$$cR^{-1} \bmod p \equiv c_1 + \left(c_0 R^{-1} \bmod p\right) \pmod{p}. \tag{3}$$

The two identities are dual under the simultaneous exchanges $R \leftrightarrow R^{-1}$ and $c_0 \leftrightarrow c_1$.

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Proof. From (1), $c \equiv c_1 R + c_0 \pmod p$, giving (2). Multiplying both sides of (1) by R^{-1} and reducing modulo p gives (3). The duality assertion is immediate from the symmetric form. □

Remark 2.2. The decomposition (1) is an exact integer identity. Barrett reduction realizes (2) by processing limbs from the most significant digit downward; Montgomery reduction realizes (3) by processing limbs from the least significant digit upward.

3. Multi-precision reduction in $n^2 + n$ digit multiplications

Write a $2n$-digit product as

$$c = \sum_{i=0}^{2n-1} c_i' r^i, \qquad 0 \le c < R^2.$$

Montgomery scan [4]. For $j = 0, 1, \ldots, n-1$, let $\tilde{c}_j$ denote the current least significant digit after j prior reductions. Choose $m_j \in \{0, \ldots, r-1\}$ satisfying

$$\tilde{c}_j + m_j p \equiv 0 \pmod r, \qquad \text{i.e.} \quad m_j = \mu_r \tilde{c}_j \bmod r,$$

where $\mu_r = -p^{-1} \bmod r$ is precomputed. Add $m_j p$ to the current sum, then divide by r (the least significant digit is now zero). After n rounds the sum is congruent to cR^{-1} modulo p; for an input with $0 \le c < pR$ it satisfies $0 \le S < 2p$, so one conditional subtraction of p normalizes (Proposition 6.1). Each round uses one digit multiplication for m_j and n digit multiplications to form $m_j p$, hence $n^2 + n$ digit multiplications in total. This is Montgomery's multiprecision REDC.

Barrett scan [2]. Choose the least integer $k \ge 1$ with $kp > R$; this ensures the quotient digit m_j remains a single digit. For $j = 0, 1, \ldots, n-1$, let $\tilde{c}_j$ be the current most significant digit and set

$$m_j = \lfloor \hat{\mu}_r \, \tilde{c}_j / r \rfloor, \qquad \hat{\mu}_r = \lfloor rR/(kp) \rfloor,$$

so that m_j is the largest single digit such that subtracting $m_j kp$ (left-aligned) from the current sum is nonnegative. Since m_j uses only $\tilde{c}_j$, the subtraction may remove one fewer copy of kp than would zero that digit, leaving a most significant digit of 1 rather than 0; record this bit and discard the digit. After n rounds the low part is $\alpha < R$, and the n recorded bits determine a correction β from a precomputed set of 2^n values, giving $\alpha + \beta$; a final addition and at most k subtractions of p normalize. The digit cost is again $n^2 + n$ digit multiplications.

Remark 3.1 (Why Montgomery is preferred on CPUs)**.** In the Montgomery scan, carries from adding $m_j p$ propagate to more significant limbs that are yet to be processed; overflows are causal. In the Barrett scan, the dual ordering causes non-causal overflows, which must be deferred to post-processing via the β-correction, introducing conditional work. In contrast, Barrett reduction operates directly in the standard residue class and avoids Montgomery-form conversions; Section 8 shows that in several important settings Montgomery reduction can nevertheless be used transparently, with the same outward behavior as Barrett.

4. Multi-precision reduction in $n^2 + 1$ digit multiplications

This section gives the basic *iterative* form of EMR; Section 5 presents two further variants. In each round of the standard Montgomery scan, the current least significant digit $\tilde{c}$ is eliminated using the precomputed μ_r and a multiplication by p, at a cost of $n+1$ digit multiplications. We observe that the same per-round reduction $\tilde{c}\, r^{-1} \bmod p$ can instead be computed as the integer product $\tilde{c} \cdot \rho_1$, where $\rho_1 = r^{-1} \bmod p$ is precomputed, at a cost of only n digit multiplications.

Proposition 4.1. *For every digit $\tilde{c} \in \{0, \ldots, r-1\}$,*

$$\tilde{c} \cdot r^{-1} \equiv \tilde{c} \cdot \rho_1 \pmod p, \qquad \rho_1 = r^{-1} \bmod p.$$

The integer product $\tilde{c}\,\rho_1$ satisfies $0 \le \tilde{c}\,\rho_1 < rR$ and therefore has at most $n+1$ digits in base r.

Proof. The congruence is immediate from $\rho_1 \equiv r^{-1} \pmod p$. The bound holds because $0 \le \tilde{c} < r$ and $0 \le \rho_1 < p < R = r^n$, so $\tilde{c}\,\rho_1 < r \cdot r^n = r^{n+1}$. □

Given a current sum S with least significant digit $\tilde{c}$, write $S = \tilde{c} + rS'$ where $S' = \lfloor S/r \rfloor$. Then $S \cdot r^{-1} \equiv S' + \tilde{c} \cdot r^{-1} \equiv S' + \tilde{c}\,\rho_1 \pmod p$. Thus each round may be replaced by: remove $\tilde{c}$, shift down by r, and add the $(n{+}1)$-digit product $\tilde{c}\,\rho_1$. Because this product has $n+1$ digits, the shortcut is applicable while the remaining sum still has at least $n+1$ digits for the carry to propagate into, that is, for the first $n-1$ rounds.

Algorithm 4.2 (Iterative EMR)**.** Given $c = \sum_{i=0}^{2n-1} c'_i r^i$ with $0 \le c < pR$, precomputed $\rho_1 = r^{-1} \bmod p$, and $\mu_r = -p^{-1} \bmod r$:

(1) Let $S = c$.
(2) **For** $j = 0, 1, \ldots, n-2$: let $\tilde{c}_j = S \bmod r$. Set $S \leftarrow \lfloor S/r \rfloor + \tilde{c}_j\, \rho_1$.
(3) **Final round:** let $m = \mu_r (S \bmod r) \bmod r$. Set $S \leftarrow (S + mp)/r$. Subtract p as needed to normalize; see Section 6 for the bound.

Theorem 4.3. *Algorithm 4.2 returns $cR^{-1} \bmod p$ using n^2+1 digit multiplications.*

Proof. At each step $j \in \{0, \ldots, n-2\}$ the current sum S with least significant digit $\tilde{c}_j$ is replaced by $\lfloor S/r \rfloor + \tilde{c}_j \rho_1 \equiv S r^{-1} \pmod p$. After $n-1$ such steps the sum is congruent to $c r^{-(n-1)} \pmod p$. The final standard round multiplies by r^{-1} once more, giving $c r^{-n} = cR^{-1} \pmod p$. Each of the first $n-1$ steps costs n digit multiplications (one digit times n-digit ρ_1); the final round costs $n+1$. Total: $(n-1)n + (n+1) = n^2 + 1$. □

Remark 4.4. For $n = 1$ the improvement vanishes: both algorithms use two digit multiplications. For example, with $n = 4$ (a 256-bit prime with 64-bit digits) the saving is three digit multiplications per reduction.

5. Parallel and logarithmic variants

Generalizing Proposition 4.1, define $\rho_i = r^{-i} \bmod p$ for $i = 1, \ldots, n-1$.

Parallel EMR eliminates all $n-1$ low limbs in a single pass:

$$c r^{-(n-1)} \bmod p \;\equiv\; \sum_{i=n-1}^{2n-1} c'_i\, r^{i-(n-1)} \;+\; \sum_{i=0}^{n-2} c'_i\, \rho_{n-1-i} \pmod p, \tag{4}$$

where the first sum is $\lfloor c/r^{n-1} \rfloor$ and the second captures the fractional part modulo p. The $n-1$ products $c'_i\, \rho_{n-1-i}$ in the second sum are mutually independent, and one standard Montgomery round on the result completes the reduction in $(n-1)n + (n+1) = n^2 + 1$ digit multiplications, the same count as the iterative form.

Log EMR instead reduces the limbs in $\lceil \log_2 n \rceil$ passes, halving the number of limbs treated concurrently at each pass and finishing with one standard round; it retains the n^2+1 count while trading precomputed constants for shorter dependency chains. The three variants share the digit-multiplication count but differ sharply in their normalization cost, which we analyze in Section 6.

Remark 5.1 (EBR)**.** The Barrett scan of Section 3 admits the same optimization by duality [2], utilizing $\hat{\rho}_i = Rr^i \bmod p$ in the same, yet dual, manner.

6. Overflow and normalization

Throughout this section we take

$$0 \le c < pR,$$

as holds for a product of residues $c = ab$ with $0 \le a, b < p$. Each variant returns a value $S \equiv cR^{-1} \pmod p$; writing $x = cR^{-1} \bmod p \in [0, p)$ we have $S = x + \delta p$, and $\delta = \lfloor S/p \rfloor$ is the number of conditional subtractions of p needed to bring S into $[0, p)$. We bound δ for each variant. Recall $\rho_k = r^{-k} \bmod p \in [1, p-1]$ and set

$$\Sigma_m := \sum_{k=1}^{m} \rho_k, \qquad 1 \le \Sigma_m \le m(p-1).$$

Proposition 6.1 (Baseline)**.** *Standard Montgomery reduction (Section 3) returns $S < c/R + p < 2p$, so $\delta \le 1$: the result is x or $x + p$.*

Proof. With $S_{j+1} = (S_j + m_j p)/r$ and $0 \le m_j \le r-1$, induction gives $S_n = (c + pM)/R$ where $M = \sum_{j=0}^{n-1} m_j r^j \le (r-1)\frac{r^n - 1}{r-1} = R - 1$. Hence $S_n \le c/R + p(R-1)/R < c/R + p < 2p$. □

The remaining variants are instances of one scheme. A *reduction pass of width* b replaces a value $S = \sum_i s_i r^i$ by

$$\Pi_b(S) := \left\lfloor S/r^b \right\rfloor + \sum_{i=0}^{b-1} s_i\, \rho_{b-i}.$$

Lemma 6.2. *$\Pi_b(S) \equiv Sr^{-b} \pmod p$, and $\lfloor S/r^b \rfloor \le \Pi_b(S) \le S/r^b + (r-1)\Sigma_b$.*

Proof. Since $\lfloor S/r^b \rfloor = (S - \sum_{i<b} s_i r^i)/r^b \equiv (S - \sum_{i<b} s_i r^i) r^{-b} \pmod p$ and $\sum_{i<b} s_i \rho_{b-i} \equiv \sum_{i<b} s_i r^{i-b} = (\sum_{i<b} s_i r^i) r^{-b} \pmod p$, their sum is $Sr^{-b} \pmod p$. For the magnitude, $0 \le s_i \le r-1$ and $\lfloor S/r^b \rfloor \le S/r^b$ give $\Pi_b(S) \le S/r^b + (r-1)\sum_{i<b} \rho_{b-i} = S/r^b + (r-1)\Sigma_b$, while $\Pi_b(S) \ge \lfloor S/r^b \rfloor$. □

A *schedule* is a composition $b_1 + \cdots + b_T = n-1$ of the $n-1$ pre-final reductions; put $B_t = b_1 + \cdots + b_t$. EMR applies $\Pi_{b_1}, \ldots, \Pi_{b_T}$ to c and finishes with one standard Montgomery round.

Theorem 6.3. *For any schedule, the EMR output satisfies $S \equiv cR^{-1} \pmod p$ and*

$$S \;<\; \frac{c}{R} \;+\; \frac{r-1}{r} \sum_{t=1}^{T} \frac{\Sigma_{b_t}}{r^{(n-1)-B_t}} \;+\; p.$$

Proof. Let $S^{(0)} = c$ and $S^{(t)} = \Pi_{b_t}(S^{(t-1)})$. Lemma 6.2 gives $S^{(t)} \le S^{(t-1)}/r^{b_t} + (r-1)\Sigma_{b_t}$ and $S^{(t)} \equiv S^{(t-1)} r^{-b_t} \pmod p$, so $S^{(T)} \equiv cr^{-(n-1)} \pmod p$. Unrolling the inequality and using $B_T = n-1$,

$$S^{(T)} \le \frac{c}{r^{n-1}} + (r-1) \sum_{t=1}^{T} \frac{\Sigma_{b_t}}{r^{(n-1)-B_t}}.$$

The final standard round yields $S = (S^{(T)} + mp)/r \le S^{(T)}/r + (1-1/r)p < S^{(T)}/r + p$ with $S \equiv cr^{-n} = cR^{-1} \pmod p$. Dividing the displayed bound by r gives the claim. □

Corollary 6.4 (Iterative EMR)**.** *For the schedule $(1, \ldots, 1)$,*

$$S \;<\; \frac{c}{R} + \rho_1\bigl(1 - r^{-(n-1)}\bigr) + p.$$

Writing $T_{\mathrm{iter}} := \rho_1(1 - r^{-(n-1)})$, we have

$$\delta \;\le\; \left\lfloor \frac{c}{Rp} + \frac{T_{\mathrm{iter}}}{p} + 1 \right\rfloor \;\le\; 2$$

whenever $0 \le c < pR$ and $\rho_1 < p$; in particular $\delta \le 2$ for all n. Moreover $\delta \le 1$ whenever $T_{\mathrm{iter}} < p - c/R$.

Proof. Here $\Sigma_{b_t} = \rho_1$ and $B_t = t$, so $\sum_{t=1}^{n-1} \rho_1 r^{-(n-1-t)} = \rho_1 \sum_{s=0}^{n-2} r^{-s} = \rho_1 \frac{r}{r-1}\bigl(1 - r^{-(n-1)}\bigr)$; multiplying by $(r-1)/r$ leaves T_{iter}. Dividing the overflow bound by p gives $S/p < c/(Rp) + T_{\mathrm{iter}}/p + 1$. Since $c/R < p$ and $T_{\mathrm{iter}} < \rho_1 < p$, the bracket is below 3, so $\delta \le 2$. If $T_{\mathrm{iter}} < p - c/R$, then $S/p < 2$ and $\delta \le 1$. □

Corollary 6.5 (Parallel EMR)**.** *For the schedule $(n-1)$,*

$$S \;<\; \frac{c}{R} + \frac{r-1}{r}\,\Sigma_{n-1} + p.$$

Writing $T_{\mathrm{par}} := \frac{r-1}{r}\,\Sigma_{n-1}$, we have

$$\delta \;\le\; \left\lfloor \frac{c}{Rp} + \frac{T_{\mathrm{par}}}{p} + 1 \right\rfloor \;\le\; n$$

whenever $0 \le c < pR$; in particular $\delta \le n$ for all $n \ge 2$. Moreover $\delta \le 1$ whenever $T_{\mathrm{par}} < p - c/R$.

Proof. The single pass has $B_1 = n-1$, giving the overflow bound. Dividing by p yields $S/p < c/(Rp) + T_{\mathrm{par}}/p + 1$. Using $c/R < p$ and $\Sigma_{n-1} \le (n-1)(p-1)$ gives $T_{\mathrm{par}}/p < \frac{r-1}{r}(n-1)$, hence $S/p < 2 + \frac{r-1}{r}(n-1) < n+1$ and $\delta \le n$. If $T_{\mathrm{par}} < p - c/R$, then $S/p < 2$ and $\delta \le 1$. □

Corollary 6.6 (Log EMR)**.** *For the halving schedule* $D_0 = n-1$, $b_t = \lceil D_{t-1}/2\rceil$, $D_t = \lfloor D_{t-1}/2\rfloor$,

$$S \;<\; \frac{c}{R} + \frac{r}{r-1}\,p + p.$$

Writing $T_{\log} := \frac{r}{r-1}\,p$, *we have*

$$\delta \;\le\; \Big\lfloor \frac{c}{Rp} + \frac{T_{\log}}{p} + 1 \Big\rfloor \;=\; \Big\lfloor \frac{c}{Rp} + \frac{r}{r-1} + 1 \Big\rfloor \;\le\; 3$$

whenever $0 \le c < pR$*; in particular* $\delta \le 3$ *for all* $n \ge 2$ *and* $r \ge 2$*, independently of* n*. Moreover* $\delta \le 1$ *whenever* $T_{\log} < p - c/R$.

Proof. Here $(n-1) - B_t = D_t$, a strictly decreasing sequence of nonnegative integers, hence the D_t are distinct; and $b_t = \lceil D_{t-1}/2\rceil \le D_t + 1$, so $\Sigma_{b_t} \le b_t(p-1) < (D_t+1)p$. Therefore, summing over distinct D_t,

$$\sum_t \frac{\Sigma_{b_t}}{r^{D_t}} < p\sum_{d=0}^{\infty}(d+1)r^{-d} = \frac{r^2}{(r-1)^2}\,p,$$

and multiplying by $(r-1)/r$ gives $T_{\log}/p = r/(r-1)$. Adding $c/R < p$ yields $S/p < c/(Rp) + r/(r-1) + 1 \le 2 + r/(r-1) < 4$, so $\delta \le 3$. If $T_{\log} < p - c/R$, then $S/p < 2$ and $\delta \le 1$. ☐

Remark 6.7 (Iterative EMR on standard 256-bit moduli)**.** Fix $w = 64$ and $n = 4$ limbs ($R = 2^{256}$) and apply iterative EMR (Corollary 6.4). The four moduli below are the base and scalar primes for P-256 [5] and secp256k1 [7]; each is a 256-bit prime with $p < R$. The table lists $\rho_1 = r^{-1} \bmod p$, the δ bound from Corollary 6.4 at a product $c < p^2$ and at a Montgomery input $c < pR$, and the offset $R - p$.

Modulus	$R-p$	ρ_1/p	$c<p^2$	$c<pR$
P-256 base	224 bits	$\ll 1$	$\delta \le 1$	$\delta \le 2$
P-256 scalar	224 bits	0.80	$\delta \le 2$	$\delta \le 2$
secp256k1 base	$2^{32}+977$	0.84	$\delta \le 2$	$\delta \le 2$
secp256k1 scalar	129 bits	0.29	$\delta \le 2$	$\delta \le 2$

Only the P-256 base modulus satisfies $T_{\text{iter}} < p - c/R$ at $c < p^2$, so a single conditional subtraction suffices there; the generic bound $\delta \le 2$ is tight for the other three moduli.

Remark 6.8 (The non-causal overflow of EBR)**.** The dual scheme, EBR (Remark 5.1), has a structurally different overflow. In Theorem 6.3 the correction injected by pass t is attenuated by the factor $r^{-((n-1)-B_t)} \le 1$ supplied by the divisions of all later passes; this geometric contraction is what bounds the Montgomery-side overflow in Corollaries 6.4–6.6. Under the duality $R \leftrightarrow R^{-1}$ with the scan reversed from least to most significant, the homologous factor for EBR is $r^{+((n-1)-B_t)} \ge 1$: corrections fall in already-processed high-order positions, where no later division acts, and are therefore *amplified* rather than contracted. The excess thus cannot be absorbed causally; it must be removed afterwards through the precomputed β-correction together with up to $k = \lceil (R+1)/p\rceil$ residual copies of p. This asymmetry, not any difference in the digit-multiplication count, is why EMR, and not EBR, is the practical algorithm.

7. Implementation

Contents TBD.

8. Applications

The EMR replaces the reduction loop in Montgomery-form modular multiplication. Standard-form multiplication by a constant K is available by storing $RK \bmod p$ instead of K: the multiplier returns $R^{-1}\cdot RK\cdot x = Kx$, eliminating Montgomery-form conversions. Two important cryptographic primitives can exploit this directly.

NTT and arithmetic hash functions. In a number-theoretic transform all multiplications are by precomputed twiddle factors, and in constant-matrix hash functions such as Poseidon2 [3] all field multiplications are by constant matrix entries. Storing each constant K as $RK \bmod p$ makes every such multiplication transparent: a Montgomery multiplier delivers Kx with no extra conversion.

Projective elliptic-curve addition. The complete projective addition formula of Renes, Costello, and Batina [6] produces an output (X_o, Y_o, Z_o) from inputs (X_1, Y_1, Z_1) and (X_2, Y_2, Z_2). If each base-field multiplier introduces a factor R^{-1}, the output is scaled by a uniform factor R^{-3}. Since the affine representation normalizes the x and y coordinates by Z, this common factor cancels and the result is correct without any conversion.

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